

Title

Aerosol Foam Delivery of Combination Vitamin D Analog and Corticosteroid Demonstrates Superior Efficacy Compared to Ointment Formulation in Psoriasis

Abstract

Three hundred seventy-six adults with plaque-type psoriasis were randomized to receive one of four formulations: combination vitamin D analog and corticosteroid in either aerosol foam or ointment vehicle, or respective vehicles alone. Primary efficacy endpoint defined treatment success as physician-assessed clinical clearing or near-clearing with minimum two-level improvement, assessed at week 4. The aerosol foam formulation achieved treatment success in 55.3% of participants versus 42.8% receiving ointment ($p = 0.022$). Modified disease area and severity index scores demonstrated significant between-group differences favoring foam (mean difference -0.62 points, $p = 0.004$). Both active formulations produced rapid and sustained itch improvement, with no meaningful differences between delivery systems. Adverse events were minimal and comparable between groups; single case of mild application site irritation with foam. These results demonstrate enhanced therapeutic efficiency with foam-based delivery of topical combination therapy for psoriasis.